The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions

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July 2026

Abstract

Krauth and Mézard (1989) predicted that the maximum number of storable patterns in the Ising perceptron is $M_N = \alpha_\star N + o(N)$ for an explicit constant $\alpha_\star \approx 0.833$. Ding and Sun proved the matching lower bound, and Huang recently proved the matching upper bound; each result is conditional on an outstanding numerical hypothesis about an explicit low-dimensional variational problem. Their global sign clauses were supported by ordinary floating-point computations; Ding–Sun's final hypothesis also contains curvature and endpoint-derivative clauses. Huang separately used rigorous interval arithmetic for the other numerical inputs, but inherited the Ding–Sun parameter rectangle and left open the main Condition 1.3—the nonpositivity, over the whole plane, of a two-variable rate function with a degenerate point and slowly approaching tails. That unbounded optimization is the reason the capacity upper bound remained conditional.

We verify both outstanding conditions in full. Reparametrizing Huang's functional into the moment coordinates of the pair $(X, \tanh X)$ compresses the plane onto a *compact* convex body and turns the entropy term into a per-cell linear function by convex duality; certified sweeps then establish nonpositivity on the whole body away from a star-shaped region around the distinguished degenerate point. On the star itself no box enclosure can work—the tilted covariance $\nabla^2\Phi$ is nearly singular and its inverse loses an essential cancellation—so we certify concavity of a majorant along rays from that point, bounding the entropy Hessian by a certified majorant of $\mathbb{E}[ff^\top \max_{b\in L} \operatorname{sech}^2(b\cdot f)]$ over sublevel-set localizations of the dual that are themselves certified by one-dimensional sweeps; no interval enclosure of the nearly singular covariance is inverted. The same machinery, in one variable, verifies the full Ding–Sun Condition 1.2, including its strict-curvature clause at the degenerate zero and its one-sided derivative clause at the singular endpoint, and we re-establish the parameter rectangle $(\alpha_\star, q_\star, \psi_\star)$ on which both papers rest. Together these close the stated conditional reduction for the Krauth–Mézard prediction.

1 Introduction

Fix $\kappa = 0$ and let $g^1, g^2, \dots$ be independent standard Gaussian vectors in $\mathbb{R}^N$. The Ising perceptron is the random subset of the discrete cube

$$S_N^M \;=\; \big\{x \in \{-1,+1\}^N : \langle g^a, x\rangle \ge 0 \text{ for all } 1 \le a \le M\big\},$$

and its capacity is M_N/N, where M_N is the largest M with $S_N^M \neq \emptyset$. Krauth and Mézard [1] analyzed the Bernoulli-disorder version with the replica method and predicted that M_N/N converges in probability to an explicit constant $\alpha_\star \approx 0.83307860$, defined below. The prediction has been one of the standard test problems for making the cavity method rigorous ever since.

Rigorous progress converged on the constant from both sides. Ding and Sun [2] proved $\liminf_N \mathbb{P}(M_N/N \geq \alpha) > 0$ for every $\alpha < \alpha_\star$; combined with the sharp-threshold theorem of Nakajima and Sun [4], this gives the lower bound with high probability. Their general-subgaussian result extends Xu's Bernoulli-model theorem [3]; see also [6]. Huang [5] proved $\mathbb{P}(M_N/N \geq \alpha) \to 0$ for every $\alpha > \alpha_\star$. Both results are conditional. The Ding–Sun theorem assumes their full Condition 1.2: a certain function $\lambda \mapsto \mathscr{S}_\star(\lambda)$, arising as the exponential rate of a restricted second moment, is negative except at its degenerate zero $\lambda = 0$ and singular endpoint zero $\lambda = 1$, with $\mathscr{S}_\star''(0) < 0$ and $\lim_{\lambda\uparrow 1} \mathscr{S}_\star'(\lambda) > 0$. Huang's theorem assumes his Condition 1.3: a two-variable function $(\lambda_1, \lambda_2) \mapsto \mathcal{S}_\star(\lambda_1, \lambda_2)$, the rate function of a planted first moment, is nonpositive on all of $\mathbb{R}^2$, with equality at $(1, 0)$. The unresolved global sign conditions were supported only by ordinary floating-point computations and remained stated as hypotheses. Huang's second arXiv version rigorously verified all other numerical conditions in Theorem 3.6 at $\kappa = 0$, but explicitly excluded Condition 1.3 itself. The final 2025 journal version of Ding–Sun retained Condition 1.2 as a hypothesis; the original arXiv version described a check using nonrigorous numerical integration.

This paper closes the remaining gap: we verify both conditions, and re-derive the parameter rectangle

$$\alpha_\star \in [0.833078599,\ 0.833078600], \qquad q_\star \in [0.56394907949,\ 0.56394908030],$$
$$\psi_\star \in [2.5763513100,\ 2.5763513224].$$

of [2, Proposition 1.3], on which the two papers' $\kappa = 0$ threshold calculations (and ours) rest, in rigorous interval arithmetic. Consequently:

Theorem 1.1. *For the Ising perceptron at $\kappa = 0$ with Gaussian disorder, $M_N/N \to \alpha_\star$ in probability. The same holds for disorder with i.i.d. mean-zero, unit-variance subgaussian entries, in particular for ± 1 entries, by the universality theorem of [4].*

Remark 1.2 (Status)**.** *This manuscript presents a new computer-assisted proof claim. Its source-bound package has passed the included internal verifiers and a separate second-system audit, but it has not yet been independently reproduced or peer reviewed. A post-release verification round—a complete replay on the attested runtime, replays on a second independent runtime, a Lean-kernel check of the exact-arithmetic certificate layer, and symbolic and high-precision numerical audits of the transcribed formulas—is archived with the repository [7].*

The verification is computer-assisted in the sense that finitely many inequalities between explicitly defined real numbers are established by interval arithmetic (we use the ball arithmetic of Arb via python-flint, with certified adaptive quadrature where appropriate and certified fixed-grid mean-value quadrature for Huang's T evaluations). It is not, however, a matter of running an integrator over the stated conditions. Three genuinely mathematical obstructions have to be removed first, and most of this paper is devoted to them.

First, Huang's condition is a statement over the whole plane. The distinguished point $(1, 0)$ has value zero, while $\mathcal{S}_\star$ also tends to zero along degenerate directions only polynomially.

Reparametrizing by the moments of the pair $(X, \tanh X)$, $X \sim N(0, \psi_\star)$ (§5), maps the plane onto a bounded convex body and removes the tail entirely, while convex duality turns the entropy term into a per-cell linear function; the bulk of the body is then certified by an adaptive sweep (§6).

Second, both variational functions attain the value zero at degenerate maximizers, where no direct interval evaluation can certify a sign, and near Huang's maximizer no box enclosure of any Hessian of the problem can work either: the tilted covariance $\nabla^2\Phi$ is nearly singular there, and its inverse—which the curvature of the entropy term is—loses an essential cancellation in interval arithmetic. We certify concavity of a majorant along rays from the maximizer instead, bounding the entropy Hessian by an explicit matrix over sublevel-set localizations of the dual variable; no interval enclosure of the nearly singular covariance is inverted (§6). The explicit positive-definite majorant is inverted by its certified 2×2 determinant. The pinned value and gradient at the center are exact identities of the fixed point and require no numerics. For the Ding–Sun condition the degenerate zero at $\lambda = 0$ is handled by derivative and second-derivative analysis, while the singular endpoint $\lambda = 1$ uses a corrected near-one chain whose numerical inputs we certify (§4, §7).

Third, the interval evaluations themselves have to be organized so that widths do not defeat the margins, which are as small as 10^{-3} in the interior of the Ding–Sun grid and near the star boundary of the Huang sweep. Every delicate cell is evaluated in exact mean-value form—value at the center plus an enclosed gradient times the radius—so that the near-cancellations the margins depend on survive enclosure; the exact identities $\mathcal{H}'(\lambda) = -(1-q_\star) \log A(\lambda)/2$ and the explicit derivative integrals supply the gradients, and parameter-ball evaluation subsumes the manual corner analyses mechanically.

The verification programs, canonical certificates, complete raw record trees, and instructions to reproduce every number in this paper are available at [7]. Every proof-relevant geometric value is frozen as an exact rational, with the stated padding, and every certified sign comparison is between balls whose correctness is guaranteed by Arb. We follow the journal numbering of [2] and the numbering of [5], except where the Ding–Sun arXiv version is named explicitly.

2 The statements being verified

We use the normalizations of [2, 5]. With φ, Ψ the standard Gaussian density and upper tail, and $\mathcal{E} = \varphi/\Psi$, set for $q \in [0, 1)$

$$F_{1-q}(x) = \frac{\mathcal{E}}{\sqrt{1-q}}\Big(\frac{-x}{\sqrt{1-q}}\Big), \quad P(\psi) = \mathbb{E}\tanh^2(\sqrt{\psi}Z), \quad R_\alpha(q) = \alpha\,\mathbb{E}F_{1-q}(\sqrt{q}Z)^2,$$

and the Gardner free energy $\mathscr{G}(\alpha, q, \psi)$ of [2, eq. (1.4)]. The threshold $\alpha_\star$ is the root of $\mathscr{G}$ along the fixed point $q = P(\psi)$, $\psi = R_\alpha(q)$; Ding–Sun Proposition 1.3 makes this precise and pins the rectangle of §3. Huang's Condition 1.3 is $\mathcal{S}_\star \le 0$, the object of §5–6; Ding–Sun's Condition 1.2 is $\mathscr{S}_\star(\lambda) < 0$ for $\lambda \notin \{0, 1\}$ together with $\mathscr{S}''_\star(0) < 0$ and $\lim_{\lambda\uparrow 1} \mathscr{S}'_\star(\lambda) > 0$, the object of §4. §7 records which theorem consumes which verified fact.

3 The parameter rectangle

Block 1 (`block1_gardner.py`) re-proves Ding–Sun Proposition 1.3 in interval arithmetic: thirteen certified inequalities establishing (i) that R_α maps the stated q-intervals into the stated ψ-intervals; (ii) the Almeida–Thouless contraction $\sup_G \mathrm{d}P(R_\alpha)/\mathrm{d}q \le 0.96 < 1$, via $P' \le 0.08$ and $\mathrm{d}R/\mathrm{d}q \le 12$; (iii) the fixed-point sign changes $P(R(q,\alpha)) - q$ at the four corners; and (iv) $\mathscr{G}_\star(\alpha_{\mathrm{ub}}) < 0 < \mathscr{G}_\star(\alpha_{\mathrm{lb}})$. Each is a one-dimensional certified Gaussian integral (adaptive `acb.integral` plus an explicit tail bound), evaluated with α, q, ψ as balls covering their whole ranges, which subsumes Ding–Sun's manual corner analysis. This establishes $\alpha_\star \in [0.833078599, 0.833078600]$ and the accompanying $q_\star, \psi_\star$ intervals rigorously.

4 The Ding–Sun condition

Ding–Sun's Condition 1.2, on $[\lambda_{\min}, 1]$ with $\lambda_{\min} := \ell(0)$, is $\mathscr{S}_\star(\lambda) = \mathcal{H}(\lambda) + \mathcal{P}(\lambda) + \mathcal{A}(\lambda) < 0$ for $\lambda \notin \{0, 1\}$, with $\mathscr{S}_\star''(0) < 0$ and $\lim_{\lambda\uparrow 1} \mathscr{S}_\star'(\lambda) > 0$. It is one-dimensional and mirrors Huang's: a value sweep on the bulk, a local second-derivative analysis at the degenerate zero $\lambda = 0$, and a separate singular endpoint estimate at $\lambda = 1$. Since $\mathcal{A} \le 0$, with $\mathcal{A}(0) = \mathcal{A}'(0) = 0$ and $\mathcal{A}''(0) \le 0$ [2, discussion following eq. (2.35)], the certified central bound $(\mathcal{H} + \mathcal{P})'' < 0$ implies $\mathscr{S}_\star''(0) < 0$. Ding–Sun's exact endpoint calculation [2, proof of Theorem 1.4] gives the stronger statement $\mathscr{S}_\star'(\lambda) \to +\infty$ as $\lambda \uparrow 1$; it is analytic and needs no numerical certificate. For the remaining global sign assertion, it suffices to bound $\mathcal{H} + \mathcal{P}$ on the positive branch and, on the negative branch, the tilted majorant $\mathcal{H} + \mathcal{Q}$ (Ding–Sun's $s = 0.2$ shift). We build the certified evaluators (`dsfun.py`): $\mathcal{H}(\lambda) = \mathfrak{H}(A(\lambda))$ through the pair-entropy Γ and the inverse map $A(\lambda) = \ell^{-1}(\lambda)$. The computational grid is parametrized by $A(\tau) = \exp(2 \operatorname{atanh} \tau)$ and $\lambda(\tau) = \ell(A(\tau))$, with the well-conditioned $D_H(A) = (1 - m^2)\big(1 - 2/(\Delta + 1)\big)$ that removes the interval blow-up of the raw $(A^2 - 1)/(\Delta + 1)^2$ form; and $\mathcal{P}, \mathcal{Q}$ through the double integral $I_s(\lambda)$, evaluated by nested certified quadrature with mean-value corrections. The bulk cells verify. Near $\lambda = 1$ the printed constants in Lemma 8.2 and Proposition 8.4 of the Ding–Sun arXiv version do not pass interval evaluation: their respective integral values are $2.678\ldots$, $3.162\ldots$, and $-0.4447\ldots$. The following records the corrected implication chain, rather than using those constants as an unexplained numerical input.

Lemma 4.1 (Corrected near-one bound)**.** *For $\lambda = 1 - \iota$ and $0 < \iota \le 0.018$,*

$$\mathcal{H}(\lambda) + \mathcal{P}(\lambda) < \mathcal{H}(1) + \mathcal{P}(1) = -\mathscr{G}_\star(\alpha_\star) = 0.$$

At the terminal cell's upper τ-boundary, the certified lower endpoint of $\lambda(0.99)$ is $0.98665\ldots > 0.982$, so this neighborhood overlaps the bulk sweep.

Proof. Put $\iota(A) = 1 - \ell(A)$ and $A(s) = \iota^{-1}(s)$. The two positive one-dimensional integrals in Lemma 8.2 of the Ding–Sun arXiv version, including explicit $|z| \ge 9$ tails, give uniformly over the parameter rectangle

$$1 - \ell(A) \le \frac{2.679}{A} + \frac{3.17}{A^2} \le \frac{2.711}{A} \qquad (A \ge 100), \qquad 1 - \ell(100) > 0.025.$$

Since $\iota(A)$ is decreasing, $A(s) > 100$ and $A(s) \le 2.711/s$ for $0 < s \le 0.025$. Ding–Sun's exact entropy derivative identity therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}s} \mathcal{H}(1 - s) = \frac{1 - q_\star}{2} \log A(s) \le \frac{1}{2} \log \frac{2.711}{s}.$$

After integration, using $\frac{1}{2}\log(2.711)+\frac{1}{2}<0.9987$,

$$\mathcal{H}(1-\iota)-\mathcal{H}(1)\le 0.9987\,\iota+\frac{\iota}{2}\log\frac{1}{\iota}. \tag{1}$$

The rescaled double integral in Proposition 8.4 of the Ding–Sun arXiv version and its positive linear term give, on $0<\iota\le 0.025$,

$$\mathcal{P}(1-\iota)-\mathcal{P}(1)\le 0.285\,\iota-0.4446\sqrt{\iota}. \tag{2}$$

Here the envelope directions are important. The argument of $\log\Psi$ is chosen to give an upper envelope, while its positive density ratio is replaced by a certified lower envelope. Since $\log\Psi\le 0$, the product is still an upper bound. The factors $\alpha_\star\sqrt{1+\lambda}$ are likewise replaced by their lower endpoints, and deleting the complement of $[-9,9]\times[0,9]$ deletes only nonpositive mass and raises the integral. The negative-z comparison also uses the separately checked pin $\gamma_{\mathrm{ub}}<\psi_{\mathrm{ub}}$. Thus no unrecorded tail or parameter direction can reverse (2).

Combining (1)–(2) and dividing by $\sqrt{\iota}$ leaves

$$f(\iota)=\sqrt{\iota}\left(1.2837+\frac{1}{2}\log\frac{1}{\iota}\right)-0.4446.$$

Its derivative has the sign of $1.2837+\frac{1}{2}\log(1/\iota)-1>0$, and the certified endpoint evaluation is $f(0.018)<-0.00287$. Hence $f(\iota)<0$ throughout the claimed interval. Finally, the last part (a) grid cell certifies $\lambda_{\mathrm{lo}}=0.98665\ldots>0.982$, providing the stated overlap. $\square$

The neighborhood of $\lambda=0$ is handled by the derivative and second-derivative certificates described in §7.

5 Huang's condition in moment coordinates

Set $q_0=q_\star$ and $\psi_0=\psi_\star$; every computation below uses balls covering their certified intervals. Huang's Condition 1.3 asks that

$$\mathcal{S}_\star(\lambda_1,\lambda_2)=\inf_{s\ge 0}\Big\{\tfrac{1}{2}s^2\psi_0+\mathrm{ent}(\Lambda_{\lambda_1,\lambda_2})+\alpha_\star\,\mathbb{E}\log\Psi(V_s)\Big\}\le 0 \qquad \text{for all } (\lambda_1,\lambda_2)\in\mathbb{R}^2,$$

where $\Lambda_{\lambda_1,\lambda_2}(x)=\tanh(\lambda_1 x+\lambda_2\tanh x)$, $X\sim N(0,\psi_0)$, $M=\tanh X$, and the constraint term depends on the profile only through the two moments

$$a_1=\mathbb{E}[X\,\Lambda],\qquad a_2=\mathbb{E}[M\,\Lambda].$$

The plane $\mathbb{R}^2$ of (λ_1,λ_2) is unbounded. Huang's identities give value zero at the distinguished degenerate point $(1,0)$; along some unbounded degenerate directions the value also approaches zero only polynomially. Verifying a nonpositivity statement over this domain by interval arithmetic is the obstruction that stopped a direct check.

We remove it by changing coordinates. The profile $\Lambda_{\lambda_1,\lambda_2}$ is exactly the maximum-entropy profile for its own moments (this is the Lagrange condition that produced the two-parameter family in the first place), so

$$\mathrm{ent}(\Lambda_{\lambda_1,\lambda_2})=H(a_1,a_2):=\max\big\{\,\mathbb{E}\ \mathrm{ent}_2(\tfrac{1+\Lambda}{2}):\mathbb{E}[X\Lambda]=a_1,\ \mathbb{E}[M\Lambda]=a_2,\ |\Lambda|\le 1\,\big\},$$

and therefore

$$\sup_{\lambda_1,\lambda_2} \mathcal{S}_\star(\lambda_1,\lambda_2) \le \sup_{(a_1,a_2)\in K} \big[\, H(a_1,a_2) + G(a_1,a_2)\,\big], \qquad G(a_1,a_2) = \inf_{s\ge 0}\big\{ \tfrac12 s^2\psi_0 + \alpha_\star\, \mathbb{E}\log\Psi(V_s)\big\}. \tag{3}$$

The key point is that the new domain

$$K = \big\{(\mathbb{E}[X\Lambda], \mathbb{E}[M\Lambda]) : |\Lambda| \le 1\big\}$$

is the *moment body* of the pair (X, M), a compact convex set with support function $h(u,v) = \sup_{|\Lambda|\le 1} \mathbb{E}[\Lambda(uX + vM)] = \mathbb{E}|uX + vM|$. It is contained in the rectangle $|a_1| \le \mathbb{E}|X|$, $|a_2| \le \mathbb{E}|M|$. The unbounded tail of the (λ_1,λ_2) plane is compressed onto ∂K, where the profiles are saturated ($|\Lambda| = 1$) and hence $H\restriction_{\partial K} = 0$. Also $\mathbb{E}|M| < \sqrt{\mathbb{E}M^2} = \sqrt{q_0}$, so $D = \sqrt{1 - a_2^2/q_0}$ below is real and uniformly positive on K.

Two consequences make (3) checkable. First, the constraint term G depends on $a = (a_1, a_2)$ explicitly, through

$$V_s(a) = \frac{-\frac{a_2}{q_0}\sqrt{q_0}\, Z - \frac{a_1}{\psi_0}\mathbf{N}}{D} + s\mathbf{N}, \quad D = \sqrt{1 - a_2^2/q_0}, \qquad \mathbf{N} = \frac{\mathcal{E}(-\gamma Z)}{\sqrt{1-q_0}}, \quad \gamma = \sqrt{q_0/(1-q_0)},$$

where Z is standard Gaussian. For the resulting one-dimensional Gaussian integral, write

$$\mathcal{T}(a,s) := \mathbb{E}\log\Psi(V_s(a)).$$

For every fixed s,

$$G(a) \le \tfrac12 s^2\psi_0 + \alpha_\star \mathcal{T}(a,s).$$

Thus a per-cell choice of s gives a certified upper bound with no inner optimization. Second, convex duality gives, for *every* dual point (b_1, b_2),

$$H(a_1,a_2) \le \Phi(b_1,b_2) - b_1a_1 - b_2a_2, \qquad \Phi(b_1,b_2) = \mathbb{E}\log 2\cosh(b_1X + b_2M), \tag{4}$$

with equality when (b_1, b_2) is the dual of (a_1, a_2). Choosing (b_1, b_2) per cell (found by a Newton solve on the convex dual objective, then used as a fixed rational) turns the entropy into an explicit linear function of (a_1, a_2), tight on the cell.

Lemma 5.1 (Moment reduction)**.** *Write $\boldsymbol{\lambda} = (\lambda_1,\lambda_2)$ and let $a(\boldsymbol{\lambda})$ denote its two moment coordinates. With H and G as above,*

$$\mathcal{S}_\star(\boldsymbol{\lambda}) = H(a(\boldsymbol{\lambda})) + G(a(\boldsymbol{\lambda})), \qquad \sup_{\boldsymbol{\lambda}\in\mathbb{R}^2} \mathcal{S}_\star(\boldsymbol{\lambda}) \le \sup_{a\in K}(H(a) + G(a)),$$

and (4) holds for all (b_1, b_2).

Proof. Fix $\boldsymbol{\lambda} = (\lambda_1,\lambda_2)$ and write $\Lambda = \Lambda_{\lambda_1,\lambda_2}$, $a_i = a_i(\boldsymbol{\lambda})$. The constraint term of $\mathcal{S}_\star$ depends on Λ only through (a_1, a_2) (it enters $\mathcal{S}_\star(\Lambda, s)$ only via $\mathbb{E}[M\Lambda] = a_2$ and $\mathbb{E}[X\Lambda] = a_1$, in the notation of §5), so $\mathcal{S}_\star(\lambda_1,\lambda_2) = \mathrm{ent}(\Lambda) + G(a_1,a_2)$. It remains to show $\mathrm{ent}(\Lambda) = H(a_1,a_2)$, i.e. that Λ maximizes the entropy among all profiles with moments (a_1, a_2). This is the

Lagrange condition that defines Λ: maximizing $\mathbb{E}\,\mathrm{ent}_2(\frac{1+\Lambda}{2})$ over $|\Lambda| \le 1$ subject to $\mathbb{E}[X\Lambda] = a_1, \mathbb{E}[M\Lambda] = a_2$ has, by pointwise optimization of the Lagrangian $\mathrm{ent}_2(\frac{1+\Lambda}{2}) + (\lambda_1 X + \lambda_2 M)\Lambda$, the maximizer $\Lambda^* = \tanh(\lambda_1 X + \lambda_2 M) = \Lambda$. Hence $\mathrm{ent}(\Lambda) = H(a_1, a_2)$ and $\mathcal{S}_\star(\lambda_1, \lambda_2) = (H + G)(a_1(\boldsymbol{\lambda}), a_2(\boldsymbol{\lambda}))$. Taking suprema gives the stated one-sided inequality; no density assertion about the finite-parameter image is needed. For (4), weak duality (and strong duality in the interior) gives $H(a) \le \inf_b[\Phi(b) - b \cdot a]$, where the inner maximization of the Lagrangian over $|\Lambda| \le 1$ gives $\max_\Lambda[\mathrm{ent}_2(\frac{1+\Lambda}{2}) + \theta\Lambda] = \log 2\cosh\theta$ at $\theta = b_1 X + b_2 M$; the stated inequality is the weak-duality direction, valid for every b. $\square$

The ray argument below differentiates H, so it also requires a finite dual throughout the star. This does not follow merely from compactness of K; we certify a uniform interior neighborhood constructively.

Lemma 5.2 (Uniform interior neighborhood)**.** *Uniformly over the certified interval for ψ_0, let $f = (X, \tanh X)$ and $a^\star = \mathbb{E}[f \tanh X] = \nabla\Phi(1, 0)$. Then*

$$B_2(a^\star, 0.0150391982) \subset K.$$

In particular, the full interval-inflated Region-I star, whose displacement from the true $a^\star$ is at most 0.012096, lies in $\operatorname{int} K$. For every point a in that star there is a unique finite dual $b(a)$, and

$$H(a) = \Phi(b(a)) - b(a) \cdot a, \qquad \nabla H(a) = -b(a), \qquad \nabla^2 H(a) = -[\nabla^2\Phi(b(a))]^{-1}.$$

Proof. Write $M = \tanh X$ and define

$$\rho_0(x) = \mathrm{sgn}(x)(1 - |\tanh x|), \qquad \rho_1(x) = \rho_0(x)\begin{cases} 1, & |x| \le 3/4, \\ -1, & |x| > 3/4. \end{cases}$$

If $|c_0| + |c_1| \le 1$, then $\Lambda_c = M + c_0\rho_0 + c_1\rho_1$ satisfies $|\Lambda_c| \le |M| + (1 - |M|)(|c_0| + |c_1|) \le 1$. Hence $a^\star + Ac \in K$, where $A_{ij} = \mathbb{E}[f_i \rho_j]$. The certified uniform enclosure is

$$A \in \begin{pmatrix} [0.15726546516, 0.15726546571] & [-0.01079401984, -0.01079401929] \\ [0.12551630837, 0.12551630878] & [0.01084146741, 0.01084146783] \end{pmatrix}.$$

For its columns p, q, interval arithmetic gives

$$\det A \ge 0.003059813871082545, \quad \|p - q\|_2 \le 0.2034559164094782,$$
$$\|p + q\|_2 \le 0.2001182846546655.$$

The image under A of the ℓ^1 unit diamond therefore contains a Euclidean ball of radius

$$\frac{\det A}{\max\{\|p - q\|_2, \|p + q\|_2\}} > 0.01503919829455.$$

The widest long-radius angular leaf has width 0.016. Including its 10^{-8} angular padding and a 10^{-10} Arb-rounding guard, the independently enclosed sine and cosine have joint norm at most $\sqrt{1 + \sin(0.016 + 2 \cdot 10^{-8} + 10^{-10})}$. The radial padding is 10^{-12}, and the stored-center interval can be $2 \cdot 10^{-7}$ away from the true center in each coordinate. Thus every interval box on which H is differentiated is within

$$(0.012 + 10^{-12})\sqrt{1 + \sin(0.016 + 2 \cdot 10^{-8} + 10^{-10})} + 2\sqrt{2}\,10^{-7} < 0.012096$$

of $a^\star$. This is the required-radius inequality replayed by the certificate. It proves the inclusion and leaves more than 0.002943 clearance beyond the padded star.

For any such a, choose $\varepsilon > 0$ with $B_2(a, \varepsilon) \subset K$. If u is a unit vector and $t \geq 0$, then

$$\Phi(tu) - tu \cdot a \geq t\big(h_K(u) - u \cdot a\big) \geq t\varepsilon,$$

because $\log(2\cosh y) \geq |y|$ and $h_K(u) = \mathbb{E}|u \cdot f|$. Thus the dual objective is coercive. Its Hessian $\mathbb{E}[ff^\top \operatorname{sech}^2(b \cdot f)]$ is positive definite: a nonzero linear combination of X and $\tanh X$ cannot vanish almost surely. The minimizer is consequently finite and unique. Pointwise entropy duality gives the equality for H, and the inverse-function theorem gives the two displayed derivative identities. $\square$

Region II certifies, cell by cell, the inequality $\min_k[\Phi(b^k) - b^k \cdot a] + \frac{1}{2}(s^C)^2\psi_0 + \alpha_\star T_C(a) < 0$ on the cell, where T_C encloses $\mathbb{E}\log\Psi(V_{s^C})$ over the cell by a mean-value rule in (a_1, a_2). By Lemma 5.1 the first term bounds H and the rest bounds G from above on the cell, so the certified inequality gives $(H + G) < 0$ there. Cells whose closure misses K are dropped: a cell lies outside K once some direction (u, v) has $\min_{a \in \text{cell}}(ua_1 + va_2) > h(u, v)$, which is checked against a certified upper bound on $h(u, v) = \mathbb{E}|uX + vM|$ over a fan of directions.

6 The compact sweep and the degenerate point

Write $a^\star = (\psi_0(1 - q_0), q_0)$ for the image of $(\lambda_1, \lambda_2) = (1, 0)$; the fixed-point identities give $H(a^\star) + G(a^\star) = 0$, and the following verification proves global maximality. We verify (3) in two pieces.

Region II (the bulk). For every cell C of a bounded, adaptively refined grid on K away from the star region of Region I, we certify

$$\big[\Phi(b^C) - b_1^C a_1 - b_2^C a_2\big] + \tfrac{1}{2}(s^C)^2\psi_0 + \alpha_\star T_C \; < \; 0 \qquad \text{on } C,$$

where b^C, s^C are the (nonrigorous) per-cell dual and tilt and T_C is a mean-value enclosure of $\mathbb{E}\log\Psi(V_{s^C})$ over C. Cells lying outside K (detected by the support function h along a fan of directions) contribute nothing and are dropped; cells straddling ∂K are handled by pulling the dual inward, using that (4) holds for any fixed b. The sweep runs in two stages—all of K minus a coarse box around $a^\star$, then that box minus the star, bisecting to side 10^{-3} along the star boundary—and a cell is skipped only if a certified witness places the full rectangle inside the core disk or inside one common signed weak-axis cone by affine inequalities at all four corners. Both stages terminate with every leaf cell strictly negative.

Region I (the maximizer). Near the maximizer the value is genuinely 0 at $a^\star$, so no direct sign check can succeed there, and interval evaluation of any Hessian of the problem over a two-dimensional box fails for a structural reason: $\nabla^2\Phi(1, 0)$ has condition number ≈ 82 (the pair $(X, \tanh X)$ is strongly correlated), so the dual image of an a-box is stretched by a factor

≈ 220 along one direction and every box enclosure of $\nabla^2 H = -[\nabla^2\Phi]^{-1}$ loses the cancellation it needs. We work instead along rays. The certificate fixes the single exact rational vector

$$C = (-2.448324,\ 0.790403)$$

globally, and for every unit direction v uses $\dot\sigma_v = C\cdot v$ on every radial band of that ray. Thus all bands for a given v concern the same function

$$\varphi_v(t) = H(a^\star + tv) + \tfrac{1}{2}s(t)^2\psi_0 + \alpha_\star\mathcal{T}\big(a^\star + tv, s(t)\big), \qquad s(t) = s_0 + t\dot\sigma_v, \quad s_0 = \sqrt{1-q_0}.$$

The implications used by the interval program are isolated in the next lemma. This makes explicit both the matrix-order direction and the quantifiers in the dual-localization check.

Lemma 6.1 (Ray and localization certificate)**.** *Let $P \Subset \operatorname{int} K$ be one certified moment polygon, let $L \subset \mathbb{R}^2$ be a closed bounded rectangle, and let $\widehat b_1, \dots, \widehat b_m$ lie in $\operatorname{int} L$. For $a \in P$ set $F_a(b) = \Phi(b) - b\cdot a$ and*

$$S_a = \{b : F_a(b) \le \min_k F_a(\widehat b_k)\}.$$

Suppose each certified subsegment $E \subset \partial L$ and each tangent polygon P_j covering P has fixed weights $w_{E,j,k} \ge 0$, $\sum_k w_{E,j,k} = 1$, for which

$$\sum_k w_{E,j,k}\{\Phi(b) - \Phi(\widehat b_k) - (b - \widehat b_k)\cdot a\} > 0 \quad (b \in E,\ a \in P_j). \tag{5}$$

Then the entropy dual $b(a)$ lies in $\operatorname{int} L$ for every $a \in P$. Moreover, with $f = (X, \tanh X)$ define

$$B_L = \mathbb{E}\left[ff^\top \max_{b\in L}\operatorname{sech}^2(b\cdot f)\right],$$

and let $\widehat B_L \succeq B_L$ be any certified positive-definite majorant. Then, for every $v \in \mathbb{R}^2$,

$$v^\top \nabla^2 H(a) v \le -v^\top \widehat B_L^{-1} v. \tag{6}$$

Consequently, fix a unit vector v, a radius $R(v) > 0$, and the preceding global choice $\dot\sigma_v = C\cdot v$. Suppose $0 = t_0 < t_1 < \cdots < t_r = R(v)$ and that for every i there are a certified polygon $P_i \Subset \operatorname{int} K$ and a closed bounded rectangle L_i satisfying the preceding localization hypotheses, with $\{a^\star + tv : t \in [t_{i-1}, t_i]\} \subset P_i$. Assume also throughout the prefix that $a_2(t)^2 < q_0$ and $s(t) \ge 0$, and that on every segment the certificate bounds the second derivative of the nonentropy part of φ_v by $v^\top \widehat B_{L_i}^{-1} v$. Then $H + G \le 0$ on the full ray prefix $0 \le t \le R(v)$.

Proof. By Lemma 5.2, F_a has the unique minimizer $b(a)$, and $b(a) \in S_a$. At least one anchor attaining the displayed minimum belongs to $S_a \cap \operatorname{int} L$. Condition (5) implies

$$\max_k\{F_a(b) - F_a(\widehat b_k)\} \ge \sum_k w_{E,j,k}\{F_a(b) - F_a(\widehat b_k)\} > 0$$

on ∂L, so S_a misses the boundary. Convexity of S_a and a line-segment argument from its interior anchor show that $S_a \subset \operatorname{int} L$.

The actual boundary sweep proves (5) in exact mean-value form. On a segment of half-length h and midpoint b_0, its lower bound is the midpoint value minus $h\sup_E|\partial_u\Phi(b) - a_u|$.

For fixed weights this lower bound is a concave function of a (affine minus a positive multiple of an absolute value), so strict positivity at every vertex of the tangent polygon implies strict positivity throughout it. The same weights are retained over the whole boundary subsegment and polygon; they are never selected separately at different vertices.

For $b(a) \in L$,

$$\nabla^2\Phi(b(a)) \preceq B_L \preceq \widehat{B}_L.$$

These matrices are positive definite, and inversion reverses Loewner order:

$$\widehat{B}_L^{-1} \preceq B_L^{-1} \preceq [\nabla^2\Phi(b(a))]^{-1}.$$

Lemma 5.2 now gives (6). On each $[t_{i-1}, t_i]$ the asserted nonentropy bound therefore gives $\varphi_v'' \leq 0$. The radial pieces form a gapless partition of $[0, R(v)]$, and the same fixed slope $\dot{\sigma}_v = C \cdot v$ is used on every piece, so these are restrictions of one continuously differentiable function rather than unrelated annular majorants. Hence φ_v' is nonincreasing on the entire prefix and φ_v is concave there. Finally, G is the infimum over the tilt, so $H + G \leq \varphi_v$, while Huang's exact fixed-point identities give $\varphi_v(0) = \varphi_v'(0) = 0$. It follows that $\varphi_v(t) \leq 0$, and therefore $H + G \leq 0$, for every $0 \leq t \leq R(v)$. □

The matrix B_L is a one-dimensional integral whose inner maximum is a per-z corner-or-zero selection. In the positive-root branch the implementation replaces the true zero-straddling strip by the certified outer strip $|z| \leq z_q$ and uses weight one there, producing the Loewner majorant $\widehat{B}_L \succeq B_L$; in the zero-root branches $\widehat{B}_L = B_L$. The global entropy bound $B_{\mathbb{R}^2} = \mathbb{E}[ff^\top]$ settles all but a narrow angular window around the weak eigenvector; inside it the certified bounded boxes become sufficiently tight around $(1, 0)$ for $\widehat{B}_L$ to retain the sharpness needed by the certificate.

It remains to match the polar certificates to the stage-2 skips. Let $\omega(\theta)$ be angular distance to either weak-eigenvector axis. The stage-2 test uses the deliberately shrunken effective radii

$$T_{\text{eff}}(\theta) = \begin{cases} 0.012, & \omega(\theta) \leq 0.1479995, \\ 0.008, & \omega(\theta) \leq 0.4199995, \\ 0.005, & \text{otherwise.} \end{cases}$$

The stage-2 inclusion witness encloses the maximum corner radius in Arb, including the certified center uncertainty. It either accepts the rectangle inside the everywhere-valid 0.005 disk, or selects one common signed weak axis and verifies positive projection and both affine half-cone inequalities at all four corners. The cone half-angles are shrunk by 0.012 and 0.030 and, like every radius, by a further $5 \cdot 10^{-7}$. These angular shrinkages exceed the adjacent Region-I chunk half-widths 0.008 and 0.025. Convexity then puts the entire rectangle inside the corresponding shrunken disk or cone. Thus every skipped rectangle lies in the union of accepted Region-I polar cells, so the two regions have neither a geometric gap nor an unchecked boundary sliver. All of these checks use the full Ding–Sun parameter balls.

7 Assembly

The pieces combine as follows.

Upper bound. Huang's upper-bound theorem [5] reduces the capacity upper bound at $\kappa = 0$ to four numerical conditions. His interval code verified the AMP and local-concavity conditions. The parameter rectangle was inherited from Ding–Sun Proposition 1.3 and is re-verified here in §3 (Block 1). His remaining Condition 1.3 is the statement $\mathcal{S}_\star \le 0$ on $\mathbb{R}^2$. In moment coordinates (§5) the one-sided reduction shows that $\sup_K(H+G) \le 0$ is sufficient; the two sweeps of Region II (§6) certify $H + G < 0$ on K outside the certified star region around $a^\star$ (the first over K minus a coarse box, the second over that box minus the star, bisecting to side 10^{-3} along the star boundary), and Region I's ray certificates give $H+G \le 0$ on the star itself. Hence $\sup_K(H + G) = 0$, and the one-sided reduction gives Condition 1.3. Huang's upper bound $\lim_N \mathbb{P}(M_N/N \ge \alpha) = 0$ for $\alpha > \alpha_\star$ becomes unconditional.

Lower bound. Under their Condition 1.2, Ding–Sun's main theorem gives

$$\liminf_{N\to\infty} \mathbb{P}(M_N/N \ge \alpha) > 0 \qquad (\alpha < \alpha_\star).$$

Nakajima–Sun's theorem for general subgaussian disorder upgrades this to high probability; it extends Xu's Bernoulli-model result. Writing $\mathrm{PG} = \mathcal{H} + \mathcal{P}$, the global-sign clause of Condition 1.2 is $\mathscr{S}_\star(\lambda) < 0$ for $\lambda \notin \{0, 1\}$, and the covering of $[\lambda_{\min}, 1]$ follows their own decomposition: the value grids on $[0.2, 0.982]$ and $[\lambda_{\min}, -0.125]$ (part (a), our Block 3a), the derivative signs $\mathrm{PG}' < 0$ on $[0.05, 0.2]$ and $\mathrm{PG}' > 0$ on $[-0.125, -0.03]$ (part (b)), the second derivative $\mathrm{PG}'' < 0$ on $[-0.03, 0.05]$ around the degenerate zero $\lambda = 0$ where $\mathrm{PG}(0) = \mathrm{PG}'(0) = 0$ (part (c)), and the near-one analysis on $[0.982, 1)$ (Block 2); the value grid reaches beyond 0.982, so the two pieces overlap. The central certificate and $\mathcal{A}''(0) \le 0$ [2, discussion following eq. (2.35)] give the separate journal-version clause $\mathscr{S}''_\star(0) < 0$; Ding–Sun's exact calculation [2, proof of Theorem 1.4] gives $\mathscr{S}'_\star(\lambda) \to +\infty$ as $\lambda \uparrow 1$, which is stronger than their endpoint clause. Parts (b) and (c) use the exact derivative identities $\mathcal{H}' = -(1 - q_\star) \log A/2$, $\mathcal{H}'' = -(1 - q_\star)/(2A\,\ell'(A))$ and certified two-dimensional integrals for I' and I'' (with closed-form Gaussian-moment bounds for the mass outside the integration box); the interval endpoints are pinned by the certified monotone map $\lambda = \ell(A(\tau))$, exactly as in part (a). On the negative interval of part (b) the margins are uniformly thin (about 2×10^{-3}) while a ball-parameter evaluation of I' wraps the cell width by two orders of magnitude, so the cells there are certified in mean-value form: I' at the exact cell center, plus a bound on $|\mathrm{PG}''|$ times the cell radius, the latter assembled from certified point values of I'' on a fine grid together with a closed-form bound on $|I'''|$ (a degree-four Gaussian-moment estimate) that controls the values between grid points.

Disorder. Krauth and Mézard's model has ± 1 (Bernoulli) disorder; we work with Gaussian disorder throughout. Nakajima–Sun show that the sharp-threshold locations differ by $o(1)$ across the standardized subgaussian class; hence the Gaussian convergence transfers to Bernoulli disorder and the other distributions stated in Theorem 1.1.

What is machine-checked. Every numerical certificate invoked above is a finite list of ball inequalities between explicitly defined real numbers, established by Arb; the analytic reductions around them (the moment-body reduction, convex duality, the ray majorants and their pinned identities, the localization and $B_L \preceq \widehat{B}_L$ matrix arguments, and the reductions to certified quadrature) are proved on paper. The nonrigorous companion (`huang_np.py`) only selects per-cell duals, tilts, anchors, and candidate boxes, which enter the certificates as fixed rationals; no certified inequality depends on it. For Region I, the final verifier does not merely inspect the signs of stored balls: from the frozen source it recomputes every fixed-weight localization witness, every entry of $\widehat{B}_L$ and its determinant, every inverse quadratic form, and every radial curvature enclosure, then requires byte-exact packet agreement. The top-level verification also binds the complete angular and radial trees, the four-artifact Region-I/II delegation chain, and the pinned Python executable and python-flint arithmetic core.

Inventory. The final source-bound runs comprise: the parameter rectangle, 13 checks; Huang Region II stage 1, 1200 top cells and 1822 verified leaves; stage 2, 240 top cells and 1499 verified leaves; Huang Region I, 1404 band jobs, 4542 certified angular leaves, and 16543 radial pieces; Ding–Sun Block 3a, 247 top cells and 263 recursive leaves; the corrected near-one block; and Block 3b/c at $K_{\mathrm{run}} = 21/2$, with 24 positive-branch top cells (124 leaves), 331 negative-branch top cells (578 leaves), and 16 central top cells (185 leaves), supported by a certified 59-point I'' grid. The canonical JSON certificates and their complete raw record trees are included with the source.

Remark 7.1 (Soundness audit)**.** *Two implementation errors were caught during this work by independent cross-checks and corrected before the final runs; we record them because they illustrate what interval arithmetic does and does not protect against. First, an early version of the fixed-grid quadrature multiplied the derivative enclosure of the mean-value remainder by the* signed *first moment* $\int_{cell}(z-m)\varphi\,dz \approx 0$*; since the enclosed derivative varies over the cell while* $(z-m)$ *changes sign, the remainder does not factor through the signed moment, and the resulting balls were systematically too tight—detected because a 30-digit quadrature of* $\nabla\Phi(1,0)$ *fell outside them. The corrected rule handles the positive and negative parts of* $(z-m)$ *separately. Second, a sign error in* ∂_s^2 *of the tilted constraint term (+ for −) survived until a finite-difference cross-check of every ingredient of* φ_v'' *against an independent floating-point implementation. Neither error is visible to the interval machinery itself: Arb guarantees the arithmetic, not the formulas. All certificates reported here postdate both corrections.*

Funding and declarations

The author acknowledges support from the European Research Council under the European Union's Horizon Europe programme (grant 101041711), the Simons Foundation, Heights Labs, and Israel Science Foundation grants 2258/19 and 4101/25.

Most of the work reported here was done by two AI systems: Fable 5 (Anthropic) and Codex (OpenAI). Fable 5 developed the main mathematical architecture—the moment-coordinate reduction, the ray majorants and their pinned identities, the $B_L \preceq \widehat{B}_L$ bound, the sublevel-set localization, and the corrected near-one chain—and wrote the bulk of the proof code and the initial manuscript. Codex performed a separate internal re-audit of the final source and proof boundary, applied and validated the confirmed full-circle normalization fix in the Region I verifier, validated and landed the repaired Block 3a and Huang closures, completed the fresh Block 3b/c and final closure, hardened the source-bound receipt and cross-platform release gates, assembled the final reproducibility package, and audited the final manuscript and public-release readiness.

Both systems operated in autonomous goal modes: the author set the objective of resolving the outstanding conditions, and the systems worked toward that objective over extended unattended sessions, choosing approaches, implementing and running certificates, detecting and repairing errors through separate cross-checks, and iterating until the verifications closed. The author supervised the runs, reviewed the mathematics, and accepts responsibility for the final manuscript. This was not a generic application of the AI systems: it is part of a broader research project of the author developing methods for how AI systems can be used to resolve open problems in mathematics, and these methods will be released in due course.

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